

A Tale of Three Intersecting Lines

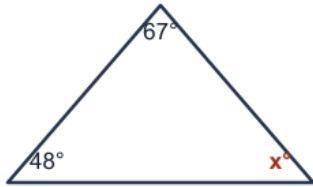
Class 7 Mathematics Worksheet | Practice Page 1 of 5

Name: _____ Date: _____

Instructions: Show your steps. Use the angle sum property, equal-angle facts, vertically opposite angles, or the exterior angle property as appropriate.

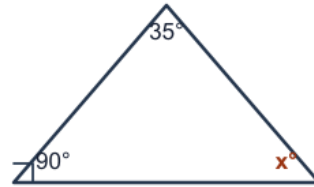
1. Find x using the angle sum property.

Diagram 1



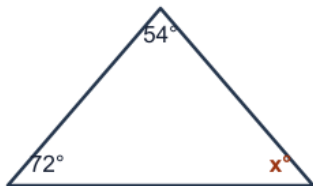
2. Find x . The square marks a right angle.

Diagram 2



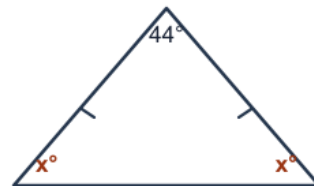
3. Find x using the angle sum property.

Diagram 3



4. The two base angles are equal. Find x .

Diagram 4

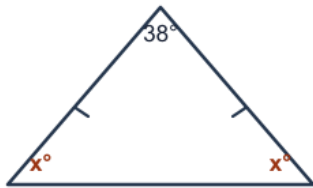


A Tale of Three Intersecting Lines

Class 7 Mathematics Worksheet | Practice Page 2 of 5

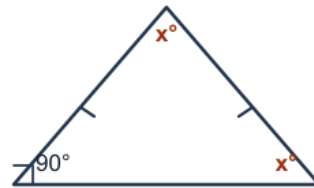
5. The two base angles are equal. Find x .

Diagram 5



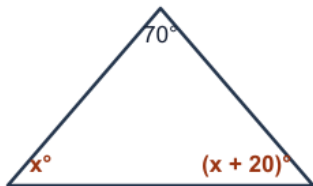
6. The two acute angles are equal. Find x .

Diagram 6



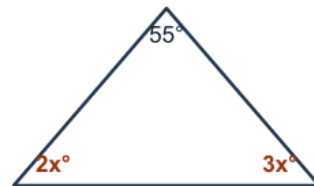
7. Find x , then find all three angles.

Diagram 7



8. Find x , then find all three angles.

Diagram 8

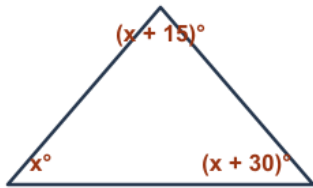


A Tale of Three Intersecting Lines

Class 7 Mathematics Worksheet | Practice Page 3 of 5

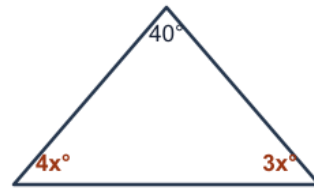
9. Find x , then find all three angles.

Diagram 9



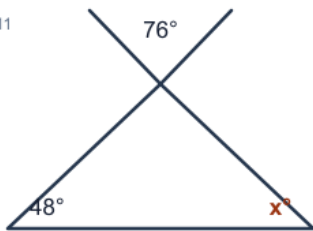
10. Find x , then find all three angles.

Diagram 10



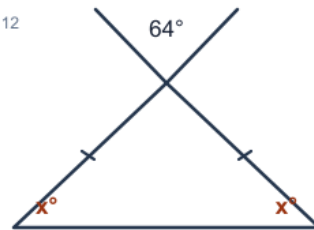
11. The sides meet as intersecting lines. Find x .

Diagram 11



12. The base angles are equal. Use the top vertical angle to find x .

Diagram 12

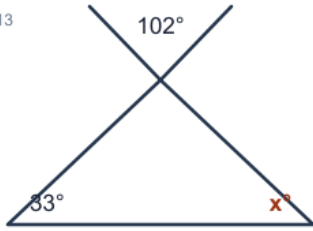


A Tale of Three Intersecting Lines

Class 7 Mathematics Worksheet | Practice Page 4 of 5

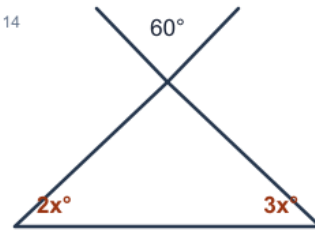
- 13.** Use vertically opposite angles and the angle sum property.

Diagram 13



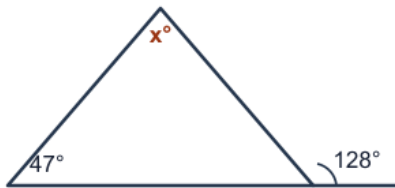
- 14.** Use the top vertical angle to find x .

Diagram 14



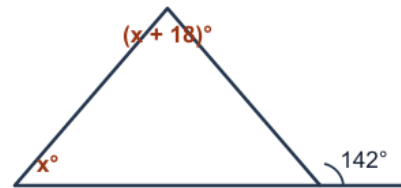
- 15.** The exterior angle is 128° . Find x .

Diagram 15



- 16.** The exterior angle is 142° . Find x and both remote interior angles.

Diagram 16

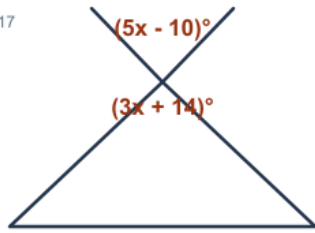


A Tale of Three Intersecting Lines

Class 7 Mathematics Worksheet | Practice Page 5 of 5

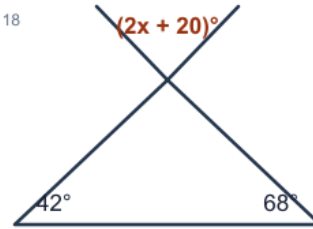
17. The marked angles are vertically opposite. Find x .

Diagram 17



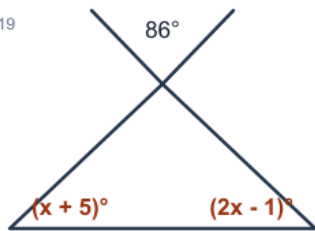
18. Use both vertically opposite angles and triangle angle sum.

Diagram 18



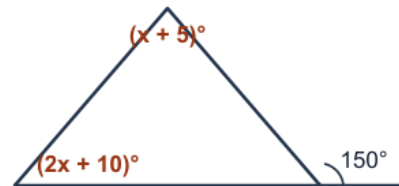
19. Find x using the top vertical angle.

Diagram 19



20. Challenge: the exterior angle is 150° . Find x and all interior angles.

Diagram 20



Answer Key

A Tale of Three Intersecting Lines | Class 7

No.	Answer	Key step
1	$x = 65^\circ$	$x = 180^\circ - (48^\circ + 67^\circ) = 65^\circ$.
2	$x = 55^\circ$	$x = 180^\circ - (90^\circ + 35^\circ) = 55^\circ$.
3	$x = 54^\circ$	$x = 180^\circ - (72^\circ + 54^\circ) = 54^\circ$.
4	$x = 68^\circ$	$2x + 44^\circ = 180^\circ$, so $x = 68^\circ$.
5	$x = 71^\circ$	$2x + 38^\circ = 180^\circ$, so $x = 71^\circ$.
6	$x = 45^\circ$	$2x + 90^\circ = 180^\circ$, so $x = 45^\circ$.
7	$x = 45^\circ$; angles: 45° , 65° , 70°	$x + (x + 20^\circ) + 70^\circ = 180^\circ$, so $x = 45^\circ$.
8	$x = 25^\circ$; angles: 50° , 75° , 55°	$2x + 3x + 55^\circ = 180^\circ$, so $x = 25^\circ$.
9	$x = 45^\circ$; angles: 45° , 60° , 75°	$x + (x + 15^\circ) + (x + 30^\circ) = 180^\circ$, so $x = 45^\circ$.
10	$x = 20^\circ$; angles: 80° , 60° , 40°	$4x + 3x + 40^\circ = 180^\circ$, so $x = 20^\circ$.
11	$x = 56^\circ$	The interior top angle is 76° (vertically opposite). $x = 180^\circ - 124^\circ = 56^\circ$.
12	$x = 58^\circ$	The interior top angle is 64° . $2x + 64^\circ = 180^\circ$, so $x = 58^\circ$.
13	$x = 45^\circ$	The interior top angle is 102° . $x = 180^\circ - 135^\circ = 45^\circ$.
14	$x = 24^\circ$; base angles: 48° and 72°	$2x + 3x + 60^\circ = 180^\circ$, so $x = 24^\circ$.
15	$x = 81^\circ$	$128^\circ = 47^\circ + x$, so $x = 81^\circ$.
16	$x = 62^\circ$; remote angles: 62° and 80°	$142^\circ = x + (x + 18^\circ)$, so $x = 62^\circ$.
17	$x = 12^\circ$; each marked angle is 50°	$5x - 10^\circ = 3x + 14^\circ$, so $x = 12^\circ$.
18	$x = 25^\circ$	Interior top angle $= 180^\circ - 42^\circ - 68^\circ = 70^\circ$. Thus $2x + 20^\circ = 70^\circ$.
19	$x = 30^\circ$; base angles: 35° and 59°	$(x + 5^\circ) + (2x - 1^\circ) + 86^\circ = 180^\circ$, so $x = 30^\circ$.
20	$x = 45^\circ$; interior angles: 100° , 50° , 30°	$150^\circ = (2x + 10^\circ) + (x + 5^\circ)$, so $x = 45^\circ$. The adjacent interior angle is 30° .